

Semiconductor Optical Single-Photon Detectors

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Abstract—Semiconductor optical single-photon detectors are solid-state photodiodes enabling the detection of individual visible photons. These are essential tools connecting the photonic and electronic domains at the quantum level. Among these semiconductor-based-detectors, single-photon avalanche diodes (SPADs) are widely implemented in the mainstream, due to their convenience at practical thermoelectrically cooled temperature, low-operating costs, and considerable high-performing detectors. Although there are challenges with the device dark count rates and after-pulsing issue, their fairly high detection efficiency (~50%) in the visible range for advancing quantum communication, cryptography, and low-light imaging.

Index Terms—avalanche diodes, single-photon detectors, photon detection efficiency, semiconductors.

I. INTRODUCTION

SINGLE-PHOTON AVALANCHE DIODES (SPADs) are photodiodes operated in a Geiger mode, so they act as single-photon detectors. A single incident photon with some probability triggers an electron avalanche that produces a “current pulse”, indicating a detection event. SPADs are practical thermoelectrically for light waves at the 950 nm to 1650 nm region in broadband communications and data transmission over optical fibers [7], over the years, various semiconductor materials have been implemented as absorbers for SPADs for different wavelength detection range.

The highest efficiency commercial devices are based on a thick, 180 μm diameter high-purity silicon absorber combined with an active quenching circuit and cooling in a single practical module. These devices offer single-photon sensitivity in the 400-1000 nm wavelength range and achieve a peak efficiency of 65% at approximately 650 nm [1].

II. OPERATIONAL PRINCIPLES OF SPADs

SPADs operate similarly to a p-n junction diode with an applied reverse-biased above the breakdown voltage [5]. In which, the bias applied to the avalanche photodiode is sufficiently large enough to cause the Markovian impact ionization gain mechanism [3]. When a photon is absorbed with energy that is higher than the bandgap, it interacts with valence electrons and elevates the electrons into the valence band. This generates electron-hole pairs in the space charge (depletion) region, the excess carriers are then subjected to strong acceleration by the electric field; the electrons are swept into the n region and the holes into the p region. In the process, these energetic carriers collide with more bound

electrons of further atoms in the material, thereby ionizing them and generating additional electron-hole pairs in a chain reaction. This avalanche flow of free charge carriers produces a measurable macroscopic current in the milliamp range [8].

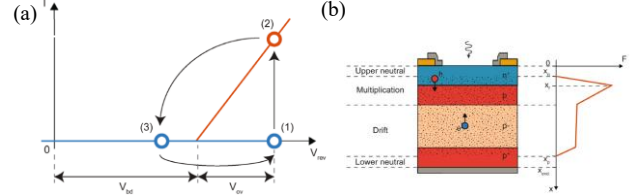


Fig. 1. Fundamental principles of a SPAD detector. a) I-V characteristic of a SPAD; V_{rev} is the reverse bias; V_{bd} is the breakdown voltage, and V_{ov} is the overvoltage. b) Basic Structure of a SPAD, with corresponding electric field F [2].

As shown in Figure 1a, a reverse bias $V_{\text{rev}} = V_{\text{bd}} + V_{\text{ov}}$ is first applied (1), no free carriers are present in the depletion region and therefore no current is flowing. In this condition, the absorption of a single photon can initiate the impact ionization process giving rise to an avalanche current (2). A quenching circuit is used to reset the bias voltage (3).

In this scenario, single-photon sensitivity depends only on the detector since the macroscopic avalanche current that is generated by a SPAD can easily overcome the noise floor of the frontend electronics. As a result, the output of the readout electronics is typically a digital pulse corresponding to the detection of a photon. When only the information about a photon impinging on the detector is retrieved, the approach is usually referred to as photon counting. Nevertheless, SPADs can also provide information about the time of arrival of each photon with picosecond precision, a feature that is exploited in the so-called photon timing.

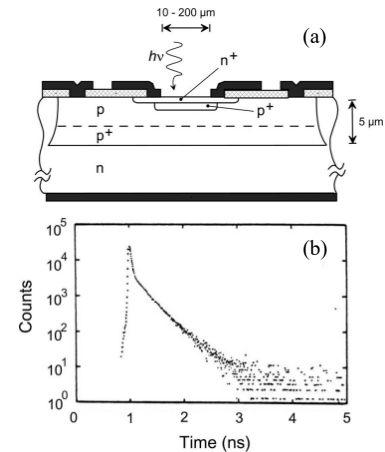


Fig. 2. Device structure and performance of a fabricated SPAD. (a) Cross-section of a double-epitaxial SPAD structure. (b) Photon arrival time distribution with this device [10].

III. CHARACTERISTICS OF SPADS

A. Photon Detection Efficiency

The PDE of a SPAD is described as the probability that a photon impinging on the active area of the detector succeeds in triggering an avalanche current that can be detected by electronics [2]. To have a macroscopic photogenerated current at the SPAD terminals, several conditions are needed: the photon must enter the device and be absorbed in its active layer; then, at least, one of the two released carriers (an electron-hole pair) must reach the high-field region and trigger the avalanche current [11-12]. The higher absorption coefficient, which depends on the wavelength and material, can increase the absorption probability of incident photons. And increasing the reverse voltage applied across SPADs' junctions can increase the triggering probability in the later condition, thus improving detection efficiency.

B. Dark Count Rate

DCR is a critical noise parameter representing the average number of dark pulses per unit time. It is associated with the avalanche current in the SPAD is triggered, when the detector is kept in perfect dark conditions and without purposely sending any photons. The DCR can originate from various mechanisms, including band-to-band thermal generation, trap-assisted thermal generation, trap-assisted tunneling, and band-to-band tunneling [11]. Each of the mechanisms shows distinguishable temperature and field dependences. This requires the band gap energy of the SPAD absorber to be larger than the thermal energy k_bT for a lower rate of thermal carrier generations. Another possible solution is to operate SPADs at lower temperatures, which can significantly reduce the thermal generation of carriers [12].

C. Afterpulsing

Afterpulsing is a secondary phenomenon associated with (DCR) that can lead to the SPAD recording of false detections. In this case, the macroscopic effect is quite different, as a train of pulses could be observed following a single-photon absorption. The physical origin of this phenomenon relies in the trapping of carriers during a previous avalanche event; such trapped carriers can be subsequently released either when the SPAD is reverse biased below or above the breakdown voltage, and in the second case a second spurious avalanche correlated to a former count is registered.

D. Dead Time

Dead time is the sum of hold-off and reset times. It represents the minimum time between two consecutive detections. The dead time also sets the maximum achievable count rate of the device, provided that the latter is not limited by other factors like power dissipation. The hold-off time is the time that the quenching circuit takes to lower the bias below the breakdown voltage. The reset time is time required to recharge the SPAD to its operating voltage.

E. Timing Jitter

Timing jitter is the timing uncertainty between the arrival of the photon at the SPAD and the detection of the avalanche

pulse by the electronic circuit [4]. This variability arises from the stochastic nature of carrier generation, transport, and avalanche multiplication processes within the SPAD [12]. The total jitter combination can be calculated from the jitter from the stochastic nature of the avalanche buildup process, the jitter contribution from the diffusion of carriers in the absorption region and the jitter from electronic noise in the readout circuitry.

F. Photon Number Resolution

PNR is important for specific applications such as quantum computation and quantum optics. SPADs may operate in free-running mode (always sensitive), or in time-gated mode (armed only during certain time windows). However, with conventional techniques of these modes, the SPAD is believed not to have the capability of PNR since avalanches always reach the saturation stage due to relatively long quenching time [9]. Operated in such a fashion, the SPAD can only resolve between zero photons and non-zero photons.

IV. QUENCHING CIRCUITS FOR SPADS

As shown in Figure 1a, after stage (2), the detector is completely blind, meaning that the absorption of another photon does not significantly change the amount of current flowing into the SPAD and therefore it cannot be detected by the electronics. And the avalanche will not stop on its own, external circuit able to quench the avalanche current by lowering the voltage below the breakdown level (3) is necessary. The quenching element can be a simple passive (PQR) in series resistor in the order of 100 k Ω or a more complex circuit such as an active quenching circuit (AQC) [2]. In the end, the SPAD must be reset to its initial bias condition (1) to make it ready to detect another photon.

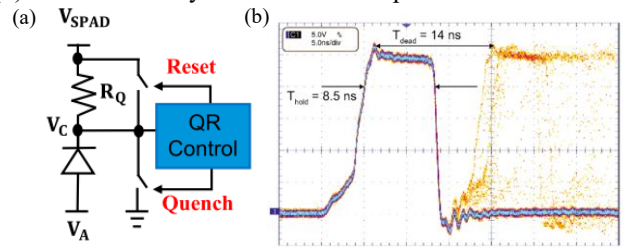


Fig. 4. Quenching circuit for a SPAD. a) Active quenching and reset circuit; R_Q is the quenching resistor; V_c is the cathode voltage; V_A is the anode voltage, and V_{SPAD} is the reverse bias voltage across the SPAD [12]. b) Voltage waveform recorded at the anode of the SPAD after the avalanche is sensed by the AQC; T_{hold} is the hold-off time, and T_{dead} is the dead time [2].

As shown in Figure 4, AQCs are employed to reduce the dead time more compared to PQRs. These circuits can shorten the dead time to a few nanoseconds by dynamically controlling the quenching and reset processes. In AQR circuits, the dead time is determined by the programmable hold-off time (which helps reduce afterpulsing by allowing trapped carriers to recombine) and reset time [12].

Passive quenching is the fundamental approach for free-running single-photon detection [9]. However, there are still two technical problems to be solved. One is long recovery time and the associated baseline shift due to avalanche pileups during recovery. The other is large afterpulse probability. The recovery problem could be overcome using a resetting circuit in practice. Thus, the afterpulsing problem remains the key challenge in implementing free-running detectors. Simply increasing the hold-off time severely limits the count rate.

V. APPLICATIONS

A. Optical Communications

In free-space optical communications, particularly in low-photon-flux regimes, SPADs provide the necessary sensitivity for reliable signal detection. Various semiconductor absorbers are selected based on the absorption coefficient and target wavelength.

The silicon platform operates in the visible spectrum (400–1000 nm). Leveraging mature CMOS fabrication, Si SPADs achieve yield with PDE > 50% and DCR < 1 kHz [10]. Below 400 nm, absorption is too shallow, while above 1.0–1.1 μm , photons are not absorbed efficiently. Engineering options include shallow and deep junctions, anti-reflection coatings, and back-illumination. Room-temperature operation is standard, with mild thermoelectric cooling sometimes being used to reduce DCR.

InGaAs/InP platform extends sensitivity into the near-infrared telecommunication range (900–1700 nm) [3]. However, these devices suffer from high DCR and after-pulsing, which restricts active areas and imposes dead times on the order of 1 ms. They are usually Peltier-cooled (-80°C to -20°C) and operate in gated mode, which lowers DCR but increases carrier trapping, further limiting maximum count rates. For example, an InGaAs Geiger-mode avalanche photodiode with a 20 mm diameter might have a 100 kHz false count rate at -50°C with a 3 ms dead time, resulting in a maximum count rate of less than 300 kHz [3].

Ge-on-Si absorbers are still in development and used only for early products as having high margins of errors in fabrication technology, to achieve sensitivity of higher wavelength spectrum (1300–1600 nm) [9]. It is suitable for integrating with silicon photonic integrated circuits. The performance is generally below InGaAs SPADs, however improving over time.

B. Quantum Technologies

SPADs can be used for the detection of entangled photons, in quantum communications with quantum key distribution (QKD), and for fundamental tests of quantum mechanics. Their single-photon sensitivity and fast timing resolution make them well suited for time-correlated single-photon counting in quantum experiments. Particularly, InGaAs/InP SPADs operating in low-frequency gating mode were foundational to the development of fiber-based quantum key distribution (QKD) [9]. These detectors were first employed in the late 1990s within auto-compensating "plug-and-play" system architectures. Subsequent years saw rapid progress in transmission distance enabled by improvements in SPAD technology. This included the demonstration of plug-and-play QKD over 40 km of fiber, one-way phase-encoding

systems over 48 km, and field tests over a 67 km installed fiber link. By utilizing designs such as a double-SPAD scheme and detectors with low dark count rates, the achievable distance was progressively extended to 100 km and ultimately to 122 km [9].

C. Other Major Implementations

SPADs can be profitably used in a wide range of other applications such as time-resolved spectroscopy; chemistry, physics, and biology; fluid velocimetry; laser ranging; optical time-domain reflectometry; single molecule detection, astronomy; distributed sensing; and optical modulators; investigations of quantum-mechanical phenomena; and studies of high field properties of semiconductors [5].

VI. CONCLUSION

The ability to generate, detect and manipulate quanta of light is the essential ground of communications applications. In this scenario, a great research effort has been devoted to the investigation of materials and solutions to design single-photon detectors able to provide a combination of high PDE, low DCR and high timing precision. Over the years, SPADs have been studied and continue to develop extensively.

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